# Module 1 Lecture - Scientific Thinking

Research Methods in Psychology

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# 1 Overview and Introduction

## 1.1 Objectives

- Understand the basis of empiricism in science, and in particular, psychology
- Understand the different but equally important roles of research producers and consumers
- Understand the fundamental structure used in the scientific process
- Appreciate norms and communication in scientific research

# 2 Basis of Empiricism

## 2.1 Psychology is a Science

- The methods of psychological research may be different from those found in chemistry, physics, or biology - but we still follow the same _____ in how we conduct our investigations.
  - After all, we are a _____!
- To be scientific, we must first be rooted in empricism, that is, to rely upon _____ and controlled observations of a phenomenon. We cannot base our decisions only in intuition, which is to make decision off of "gut feeling".
  - But, intuition *can* be part of the scientific _____, more on that later

> Discuss: Be silly - what is something that you think or believe by intuition alone, no solid evidence?

- Our scientific procedures may be _____ by confounding variables, poor ethical conduct, or limitations in design - we will discuss all of these throughout the semester
  - Scientific studies are often comprised of a balance of numerous practical choices impacting different parts of the _____ of the study.

- *Example:* Just like a chemist detailing each and every step in a successful or failed experiment, we must be equally _____ in our work as social scientists

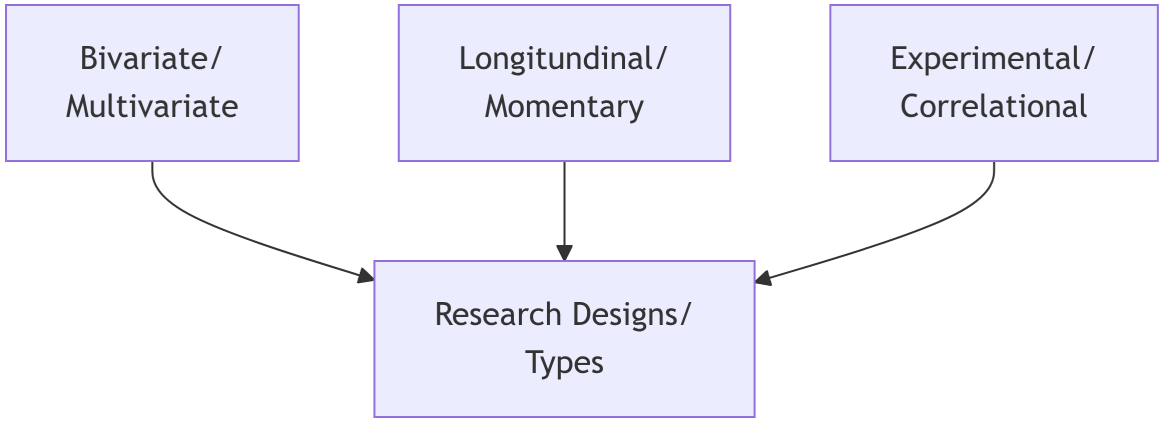
**?** Jamie is watching people shop at a store, and notices that people only buy from the racks closest to the doors. He decides to watch the store consistently for 8 hours each day over the next week, steadily tallying the number people who buy from racks in the front of the store, vs the back of the store. This process of counting and measuring is an example of [BLANK]

- (●) A) Empiricism
- ( ) B) Intuition
- ( ) C) Both of the above
- ( ) D) Neither of the above

Explanation:

## 2.2 Different Methods, Same Answers?

- There are many valid ways of _____ empirical/scientific research, many of which we will explore in this course
  - Different methods may explore the *same* overarching research question, but with different techniques, _____, and weaknesses




- It is vital that you are able to both _____ the methods of other researchers (consumer), and craft your own (producer)
  - In this course - homework and research proposal will help you strengthen both sets of skills, applying the knowledge you get in class.

# 3 Producers and Consumers

## 3.1 Research Producers

- Research Production is the process of actually _____, designing, conducting, and reporting research - using the methods we discuss in this class
  - Producing research is often a _____ of advance training in psychology (e.g., M.S., Ph.D., etc.) - and also part of your undergraduate training here at GVSU (see PSY-350 and PSY-400)
- Being an author of research establishes your _____ as an expert in a certain area or topic, and advances your analytical and writing skills
  - However, only one study does *not* make you an expert - it is the full body of your scientific and _____ work that defines your ability

> **Important**
>
> An important hallmark of good science is a healthy sense of humility - even if you do have lots of research published, it pays to be constantly trying to understand it better

- Good research is almost always goes through the process of peer review, meaning it is _____ by other individuals in that discipline.
  - Peer-review is effectively a collaboration of research producers to only publish the most _____ version of a study.
- *Example:* A scientist at a testing company (e.g., The College Board) runs analyses and reports on the metrics for the SAT over the last 5 years.

## 3.2 Research Consumers

- Applied _____ of psychology in businesses, clinics, and schools **must** understand the state of scientific literature in their respective areas. This is the process of research consumption
  - You will also be expected to be able to _____ research from journals during your training here at GVSU
- It is not enough to just read research, but also to be _____ and mindful of how "good" research is done.
  - Just because research is "peer-reviewed" does not mean it is entirely free from limitations or _____!
  - Unfortunately, some research is not properly vetted all the way; but that is why you are here!

- *Example:* A therapist applies a new, evidence-based technique for a _____

> Discuss: A professor reads about the latest education research on a new intervention to approve attention in a college classroom. She then collects data on her own students throughout the semester, while using this intervention, and reports that data in a peer-reviewed paper. Is this research consumption or research production?

### 3.3 How Producers and Consumers Compare

- Both producers and consumers play an _____ role in how science is applied:
  - Producers use _____ research designs to demonstrate real effects and relationships
  - Consumers critically _____ research findings and carefully apply them to "real-world" problems

> **Important**
>
> The same person can engage in both research production and consumption

- *Example:* An educational psychologist ( _____ ) complete a long-running study on the effectiveness of a cutting-edge intervention for disruptive behaviors in class, and a teacher ( _____ ) then implements it into their classroom management strategy.

## 4 How Scientists Work

### 4.1 Empiricism

- Scientists are empiricists that assess _____ through rigorous and systemic thinking, testing, and writing

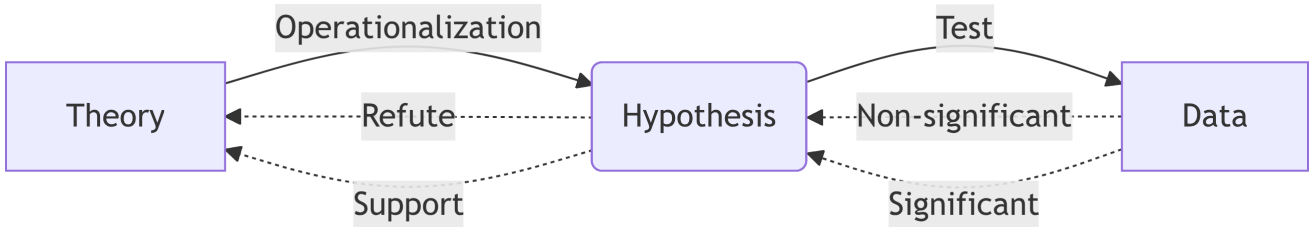
- It is not enough to just see a relationship _____; instead we must be able to observe, measure, and elicit it consistently

> **Important**
>
> Being empirical is difficult and time-consuming, but necessary.

- We may use evidence from our senses, or from measurement _____ to establish the properties and behaviors of a certain idea
  - Not all procedures for _____ or **operationalizing** are built equal, and some may be more reliable and valid than others (more in week 5!)
- *Example:* Issac Newton does not watch an apple fall from a tree just once, he drops many apples and other objects and _____ observes each.

## 4.2 The Theory-Hypothesis-Data Cycle

- Part of science is not just establishing theories and ideas, but updating them as new evidence supports or _____ with existing frameworks
  - In order to do so, we must make specific hypotheses to test, and then report on the _____ of said testing
  - Effectively we move from Theory to Hypothesis to Data, which then either refutes or supports the _____ theory




> Discuss: Do you think is acceptable to start with a hypothesis, rather than a theory? Why or why not?

## 4.3 Theories

- Theories are general statements or concepts about how a certain phenomenon is believed to _____.
  - You can also think of it as an _____
- *Example:* The _____ of gravity is a broad explanation for attraction between objects
- They are often _____ and expand over time as further information adds to and subtracts from understanding of a certain construct (Remember the Theory-Hypothesis-Data cycle from earlier!)
- These theories, oftentimes, try to describe some _____ of two or more constructs, whether that be a monkey and a figurine; a person and a treatment; a person and another person; etc.
- Most theories try to follow the rule of parsimony, that is, trying to fit the simplest-*possible* _____ for a phenomenon or observed behavior.
  - Note: not all things can be fully "simplified", but we seek the most basic and _____ explanation we can

## 4.4 Hypotheses

- Hypotheses are much more _____ statements that often serve as the foundation for any particular study. They should be pre-registered - and stated _____ to the actual commencement of the planned study.
- Making hypotheses after a study, to fit the data, is _____ (we will later touch on this issue in Week 14).
- These may be made within the context of a broader theory, but are likely to focus more concretely on a predicted outcome with _____ measures (that could be wrong!)
- *Example:* I expect that I will fall if I walk on that ice

Discuss: Get creative - come up with some other rudimentary or fun hypotheses/predictions

- Several studies, led by several hypotheses, may all contribute to the development of a grander _____

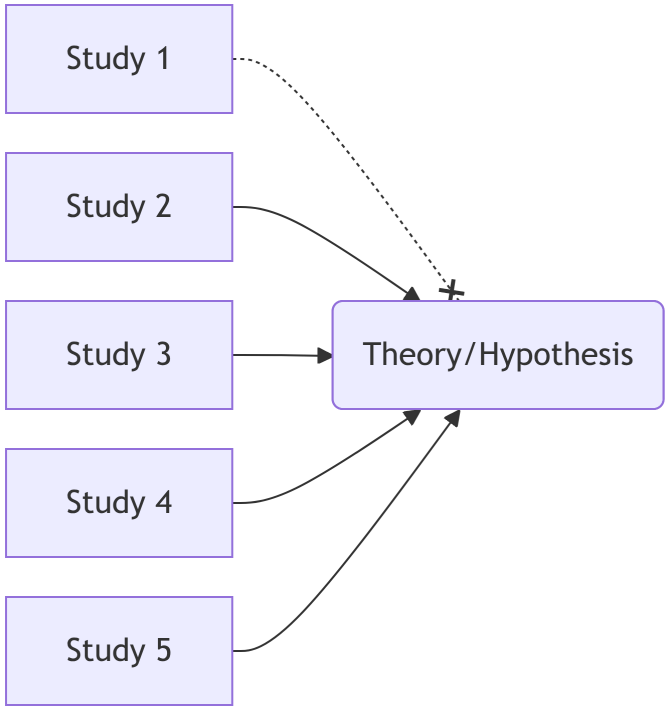
## 4.5 Data

- Data is the _____ of an experiment or study, and contains the _____ and tests that show significance or non-significance for the **hypothesis**, which aids in understanding whether the results support or refute the **theory**, respectively.
- Just like with crafting our hypothesis, we have a lot of input in how our data is treated and tested - different designs and measure will produce _____ outcomes.

Important

Data does not just appear out of thin air! We must be mindful of how it was collected and how that impacts our outcomes

## 4.6 Burden of Proof




- A singular study does not definitively _____ a certain hypothesis or theory, nor can it fully disprove these. Rather, it may add to evidence for or against a certain idea, but we must meet the burden of proof
- Example in writing: "This paper aids in understanding how CBT may be beneficial for individuals with depression. Results indicate a moderate effect of the treatment in reducing depression in the present study. Future research is needed to clarify the effect in different populations and contexts."
- Put statistically: we never *prove* or *disprove* our null hypothesis ($H_0$), we just supply evidence for or against our _____ hypothesis ($H_1$)

Discuss: Review: when something is 'statistically significant', does that suggest we should retain or reject the null hypothesis?

- Only once _____ studies have provided support for a theory,

can we say the weight of evidence is in favor of it, or that we have met the burden of proof

## 4.7 We Can be Wrong

- Falsifiability Good research must allow for our theory and/or hypothesis to be flawed or _____. If this is not accounted for, we engage in confirmation bias, or effectively choosing to only investigate for our views.
  - Recall the Theory-Hypothesis-Data flowchart from earlier. _____ is a valid outcome of new _____ !
- *Example of a non-falsifiable claim:* I may one day be a billionaire
  - It being so _____ means it is difficult to gather evidence for or against it
- It is critical that our design, statistics, and reporting make clear the _____ that a study is limited in its scope and abilities
- No one study is so _____ designed that it can account for all edge cases in a phenomenon

## 4.8 Example of Theory-Hypothesis-Data: Harlow's Monkeys

- How do we test a component of primate attachment theory? We must perform an empirical _____!
- *Example:* Harry Harlow was interested in attachment theory (broad idea set - theory) gave a young monkey two options: cling to a "wire mother" with food, or a "cloth mother" with fur and warmth. He _____ that them monkey would prefer the cloth mother (concrete prediction - hypothesis). He found that monkeys generally preferred the clothe mother (concrete outcome - _____ )
- This also shows the _____ of good research - we must be willing to be wrong (i.e., we provide the possibility that the experiment can go the "other way")
- *But,* this one study does not singularly define attachment theory, the _____ of evidence requires more studies!

## 4.9 Norms for Scientific Research

- Robert Merton proposed a set of scientific norms that can and should _____ our actions and behaviors in approaching and conducting research
- **Universalism** states that "science is for _____" and that claims are not based solely upon the expertise or stature of the scientist, but rather, their methodology and rigor
  - *Example:* an undergraduate student can perform research the same as a doctoral student, and it will be measured by its strength, not the person who made it
- **Communality** is the concept that science is done in a community and as a _____, not only a small group of individuals.
  - *Example:* Even the authors of a published paper cite many others in their writing.
- **Disinterestedness** states that we must be guided by a commitment to truth and accurate _____, not by monetary gain or pushing of a particular ideology.
  - *Example:* A prominent medical scientist publishes results about concerning side effects of a drug, despite the fact that they have stock in the pharmaceutical producing the drug.
- **Organized Skepticism** says we must commit ourselves to be critical of everything, even _____! We question things, not to simply be contrarian, but because we must understand the faults in existing knowledge.
  - *Example:* I strongly believe the MMPI to be a valid measure of personality, but I read a study that is critical of its accuracy.

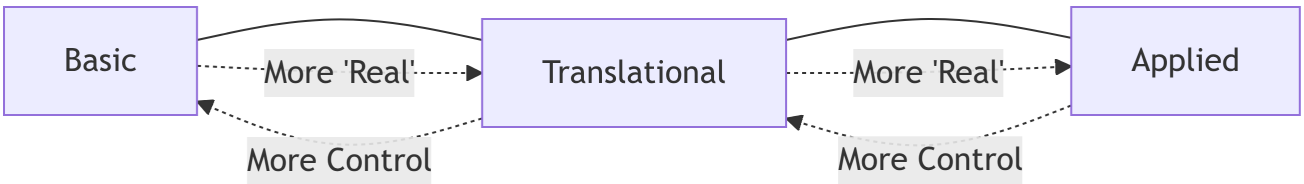
**?** James is an independent researcher, not affiliated with a school, and has produced some stellar research he wishes to share. However, journals will not accept his submission as he does not a professorship. What norm of scientific research does this violate?

- (●) A) Universalism
- ( ) B) Communality
- ( ) C) Disinterestedness
- ( ) D) Organized Skepticism

Explanation:

## 4.10 Continuum of Research Contexts

- **Basic Research** is that done for theoretical purpose to expand knowledge or _____ ideas
  - *Example:* EEG Electrodes and brain waves during a certain activity
- **Translational Research** is done in a more controlled environment, but now being applied to _____ people (or animals)
  - *Example:* Experimental study in a research lab of peoples reaction to a certain stimulus
- **Applied Research** happens more so in the "real world" where the findings from basic and translational research are applied to less-controlled _____.
  - *Example:* Retrospective study on patient outcomes after a certain treatment




- ALL forms of research here are useful and important in the _____ of well-rounded and well-supported theories!
  - As you will learn throughout the semester, certain research _____ will also lend themselves well to one of these types in particular.

**?** Bailey is conducting research on hospital cases, and reviewing data from admission to see whether certain treatments had an impact in this real life situation - which type of research from above does this best fit with?

- (●) A) Basic Research
- ( ) B) Translational Research
- ( ) C) Applied Research
- ( ) D) None of the Above

Explanation:

## 4.11 "Publicly" Available Research

- Scientists _____ primarily through publishing findings in academic journals

that use a system of editors and peer reviewers to ensure the rigor and validity of a study

Discuss: Most journals editors and peer reviewers work on a largely volunteer basis (i.e. for free). Why do you think this is and is it 'fair'?

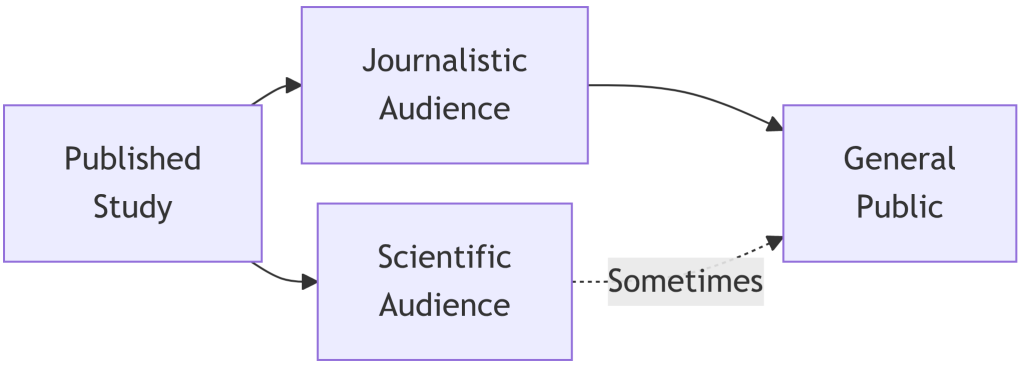
- If a paper is published, it may be _____ by future scientists in support of certain claims and arguments in papers, presentations, and reports. Or, others can also disagree and provide competing evidence in their own work.
  - A theory is not crafted from the work of just one author, but instead the synthesis of _____ different published articles
  - In psychological science we use _____ style (in its 7th edition) to reference previous scientific work
- Journals may range in quality and _____ though! We will discuss some nuances in this later in the semester
- Also, some journals may be difficult to _____, but there are many ways we may use the library resources to access the texts
  - By the end of this semester, you should be very familiar with what a peer-reviewed publication looks like.

## 4.12 Scientific Journalism

- Specialized journalists often try to bring scientific findings (published in journals) to a _____ that is more acceptable to laypeople.
  - Scientific journalisms is important work, as it can help _____ new knowledge.
- However, these writings are *not* peer-reviewed the same as the original research - and may _____, understate, or be reductive towards the "true" findings
  - It is not that journalism is _____ bad, just that we must be cautious
- When in doubt $\longrightarrow$ always go to the original publication!!

  - Note: Citing journalism about a study, rather than the study itself, may be improper _____
- There is an old saying in journalistic media: "If it bleeds, it leads" - referring to how more violent and shocking stories tend to be covered more and generate more public interest. The same thing happens in this type of _____ journalism!
  - Journalists may occasionally prioritize writing in a way that sounds "flashy".
  - *Example:* "Promising new drug shows potential to cure Alzheimer's"

📢 Discuss: Some informal homework to take home: find a headline about scientific results that seems to good to be true




# 5 Conclusion

## 5.1 Recap

- We discussed how research is rooted in empiricism, and follows a strict, scientific process, but may also come in many different forms and types
- We talked about positive norms to follow in creating good research, and how to communicate in a way that is respected and follows the importance of peer review
- We also touched on what the differences are between research consumers and producers, and how they both serve important roles in the broader use of

research

## Key Terms

**A**

**academic journals** Groups of articles published after thorough peer review 12

**B**

**burden of proof** The need to have multiple pieces of agreeing evidence to support an idea or conclusion 9

**D**

**Data** The output or results of measurement from a study/experiment 8

**E**

**empricism** The orientation towards systemic and rigorous investigation of phenomena 2

**F**

**Falsifiability** The necessity of our hypotheses to be testable and be able to be proven wrong 10

**H**

**Hypotheses** Specific predictions about two or more variable in a certain context 7

**I**

**intuition** Use of our 'gut feeling' to make decisions and determinations 2

**P**

**peer review** The process by which newly created research is audited and reviewed by other experts to asses its credibility 4
**pre-registered** Publically stating a hypothesis prior to designing a study and gathering data 7

**R**

**research consumption** When one reads, understands, and applies peer-reviewed research in practical applications 4
**Research Production** the process by which one conducts, creates, and analyze phenomena, reporting results to others 4

**rule of parsimony** A rule that states we should observe the simplest possible explanation that still maintains necessary complexity 7

**S**

**Scientific journalism** The re-writing of scientific results from peer reviewed sources into a format more accessible to the public 13

**T**

**Theories** Broad or general statements about the mechanisms of how things are related 7

**W**

**weight of evidence** The cumulative sum of several different pieces of evidence to lend towards a certain conclusion 10

*The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms*